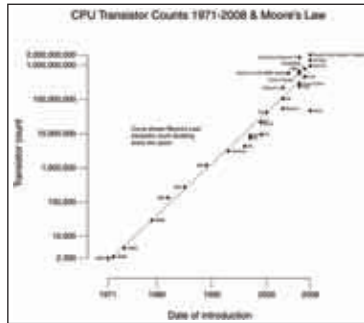


age, they will wonder why it was not until the year 1960 that anybody began seriously to move in this direction.” Without ever using the word “nanotechnology” he predicted the possibilities and potentials of nano-sized materials.

Moore's Law (1965)

Intel co-founder Gordon Moore described in Electronics magazine several trends he foresaw in the field of electronics. One trend now known as “Moore’s Law,” described the density of transistors on an integrated chip (IC) doubling every 12 months (later amended to every 2 years). Moore’s prediction turned out to be prophetic. In fact, the complexity of a chip continued to double yearly, long after 1975. The rate of doubling



Graph of Moore's Law

has only recently slowed to about every 18 months. That the basic trend Moore envisioned has continued for 50 years is to a large extent due to the semiconductor industry’s increasing reliance on nanotechnology as ICs and transistors have approached atomic dimensions.

Coining the Term (1974)

Tokyo Science University Professor Norio Taniguchi coined the term nanotechnology to describe precision machining of materials to within atomic-scale dimensional tolerances.

He used the word to refer to “production technology to get the extra high accuracy and ultra-fine dimensions, i.e. the preciseness and fineness on the order of 1 nano.”



Norio Taniguchi

Molecular Electronics (1974)

In 1974, Northwestern University Charles and Emma Morrison Professor of Chemistry, Mark A. Ratner and A. Aviram of IBM, proposed that individual molecules might exhibit the behavior of basic electronic devices, thus allowing computers to be built from the bottom up by turning individual molecules into circuit components. This hypothetical application of nanotechnology,